

Predicting α -helix secondary structure

A two-state Hidden Markov Model fit with amortized BayesFlow inference

Group [number]

[Member 1] · [Member 2] · [Member 3]



Roadmap

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The question Predict α -helix vs other from sequence

2

Forward–Backward Exact per-residue posterior as training targets

3

BayesFlow Amortized neural posterior via a sliding window

4

Insulin & real evaluation Held-out real proteins vs ground truth

5

Comparison & results BayesFlow vs exact Forward–Backward

From amino-acid sequence to α -helix

- Proteins fold; the local shape is the secondary structure (helix / sheet / coil).
- We predict one thing: is each residue an α -helix, or 'other'?
- Solving 3-D structure needs X-ray / NMR — expensive; we want it from sequence.
- Our statistical model: a two-state Hidden Markov Model.

α -helix = state “H”

“other” = E B G I T S C

Mapping decision

DSSP labels each residue with 8 states. We keep only H as helix (strict α -helix); the other 7 $\rightarrow$ other.

The generative model: a two-state HMM

- The chain always starts in “other”.
- Transitions: helix→helix 0.90, other→helix 0.05.
- So helices form ~10-residue runs, and are started rarely.
- Realistic: real data gives 0.912 and 0.041 (see next slide).

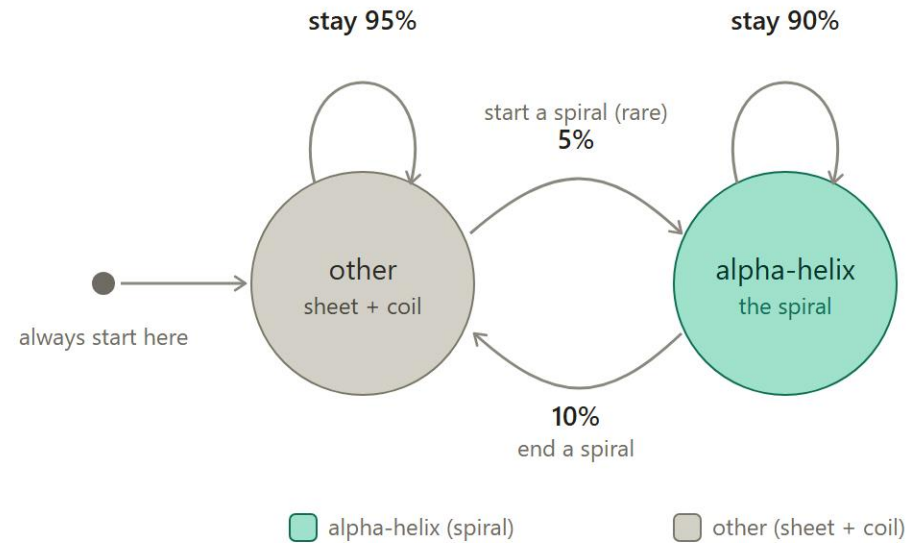


Fig 1. Two-state HMM: start rule and transition probabilities.

Emissions — and are they realistic?

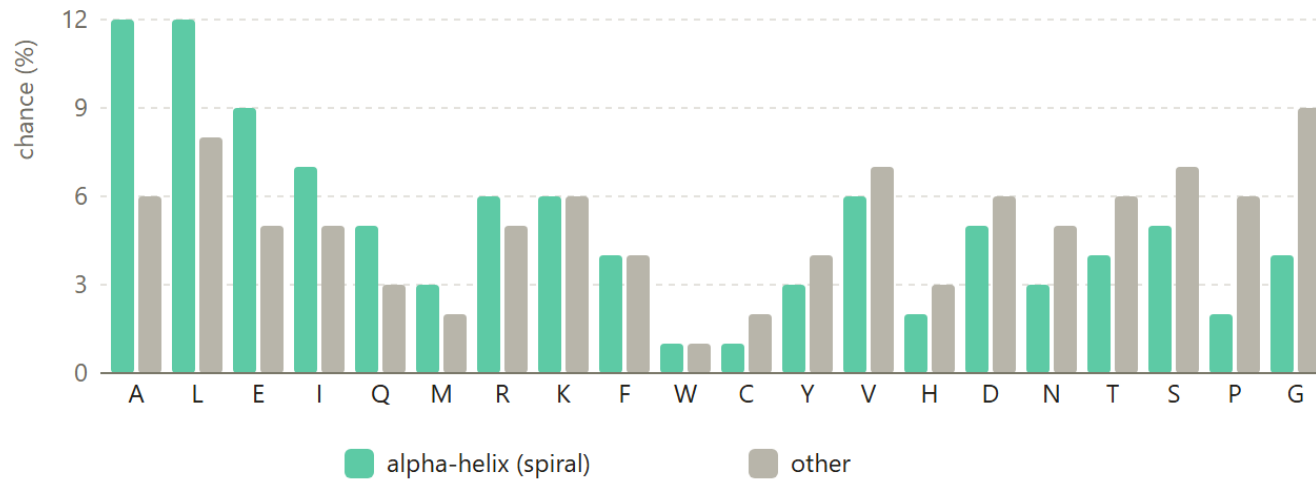


Fig 2. Per-state amino-acid emission probabilities (α -helix vs other).

Yes — checked against real PISCES data: emissions match at $r = 0.99$ (max 0.9 pp off); transitions match too ($P(h \rightarrow h)$ 0.912 vs 0.90). Both tables are empirically well-founded.

Notebooks 1–2: simulate, then get exact targets

- Simulate 100,000 chains; lengths sampled from real PISCES (median 208).
- Run Forward–Backward (hmmlearn) → exact $P(\text{helix})$ per residue = training targets.
- 24.3M residues — the exact Bayesian “answer key”.
- Checks: mean P 0.324; residue-0 = 0; recovers hidden states at the Bayes ceiling (AUC 0.79).

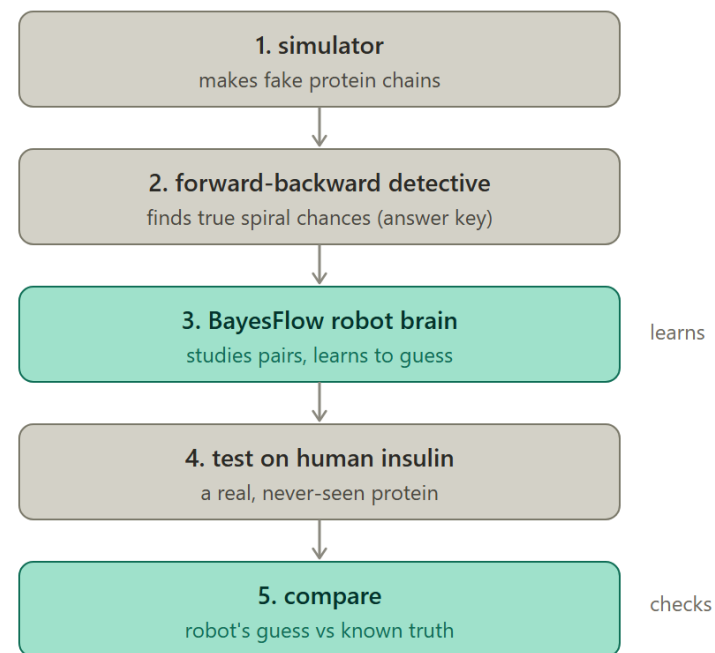


Fig 3. Full pipeline: simulator → Forward–Backward → BayesFlow → evaluation.

Turning a variable-length chain into a fixed input

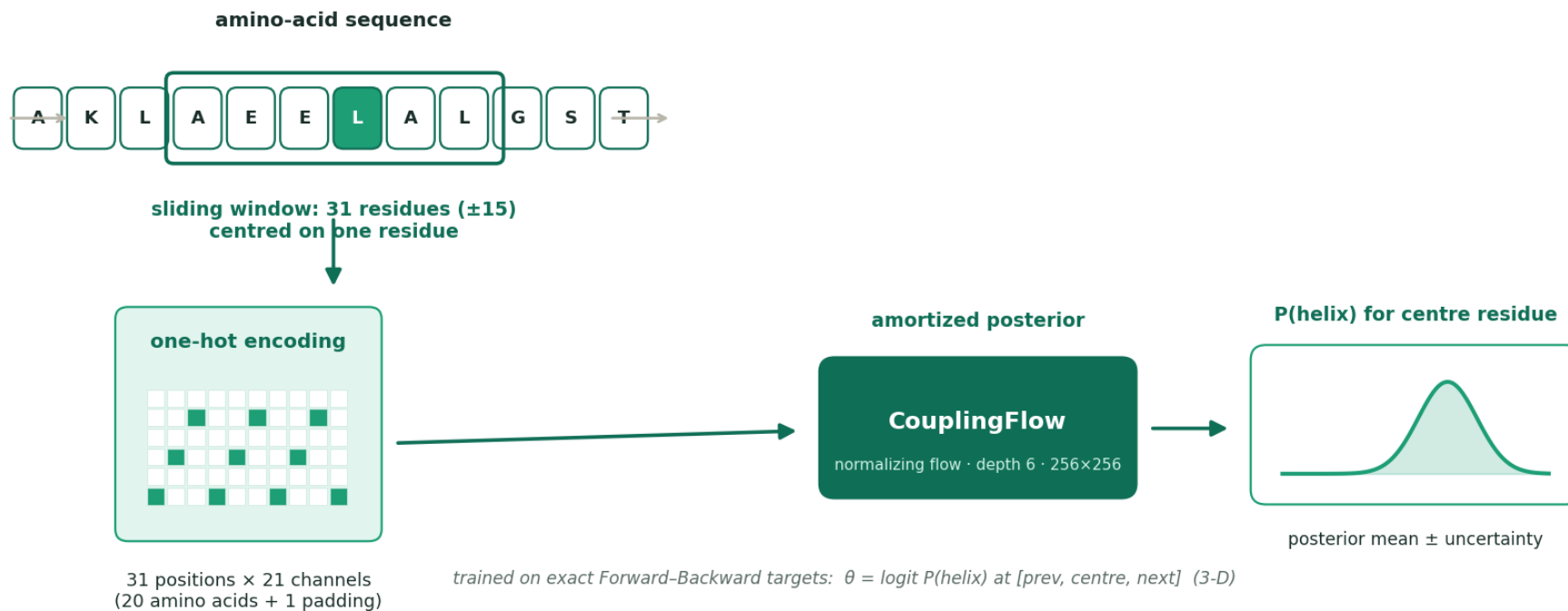


Fig 4. Sliding 31-residue window $\rightarrow$ one-hot encoding $\rightarrow$ CouplingFlow posterior.

- A flow needs a fixed-size target $\rightarrow$ predict one residue from a fixed 31-residue window (± 15), slid along the chain.
- Encode the window as one-hot: 31 $\times$ 21 channels = 20 amino acids + 1 padding (for chain ends).
- The window is only local $\rightarrow$ the answer stays uncertain $\rightarrow$ a genuine posterior (mean $\pm$ std), not a point estimate.

How we trained it

Component	Choice · why
Inference network	CouplingFlow, depth 6, 256×256 subnets
Target θ (3-D)	logit $P(\text{helix})$ at [prev, centre, next]
Conditions	flattened $31 \times 21 = 651$ one-hot window
Training data	300,000 windows from 15,000 chains
Optimizer / epochs	Adam · 30 epochs · batch 512

No leakage (verified)

Train = front block [0, 15000); validation = tail block [99500, 100000). Disjoint indices AND 0 shared sequences — asserted at runtime.

Why a window?

One trained model does inference for a chain of ANY length — just slide the window. Amortized: no re-fitting per protein.

Held-out real protein: wild-type insulin (1MSO)

- Insulin + true DSSP labels from the dataset; model trained ONLY on simulated data (unseen).
- B-chain helix is propensity-driven (L,V,E,A) → nailed, AUC 0.98.
- A-chain helix is disulfide-stabilised & Cys-rich; C disfavors helix in our table → model predicts “not helix” → AUC 0.52 (chance).
- A sequence-only HMM cannot see 3-D disulfide bonds.

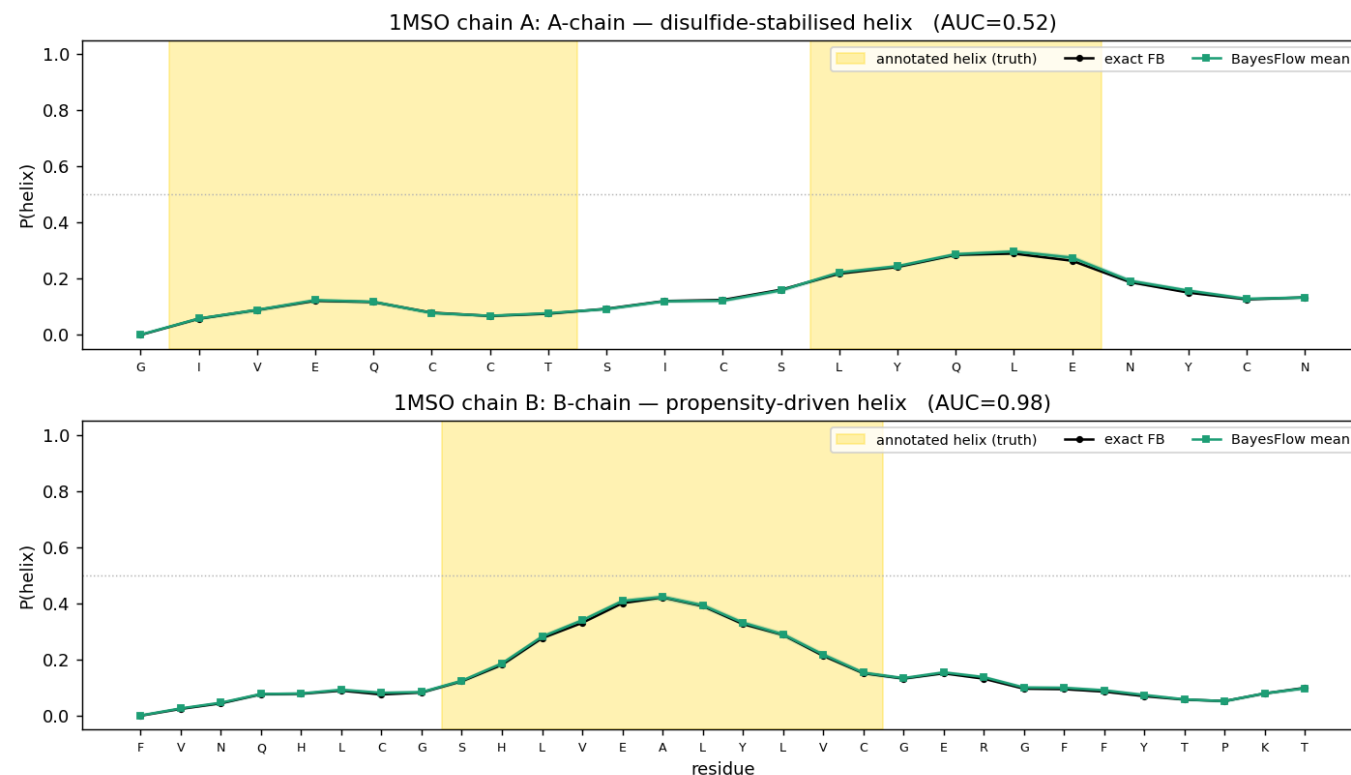


Fig 5. Insulin $P(\text{helix})$: BayesFlow (green) vs exact FB (black); gold = true helix.

All real proteins (PISCES)

- Scored all 8,994 standard-AA chains (2.2M residues).
- Per-chain AUC 0.754; BayesFlow = exact FB.
- A simple HMM ranks helices meaningfully across the messy real proteome.
- AUC > accuracy: the HMM isn't calibrated to reality → use ranking.

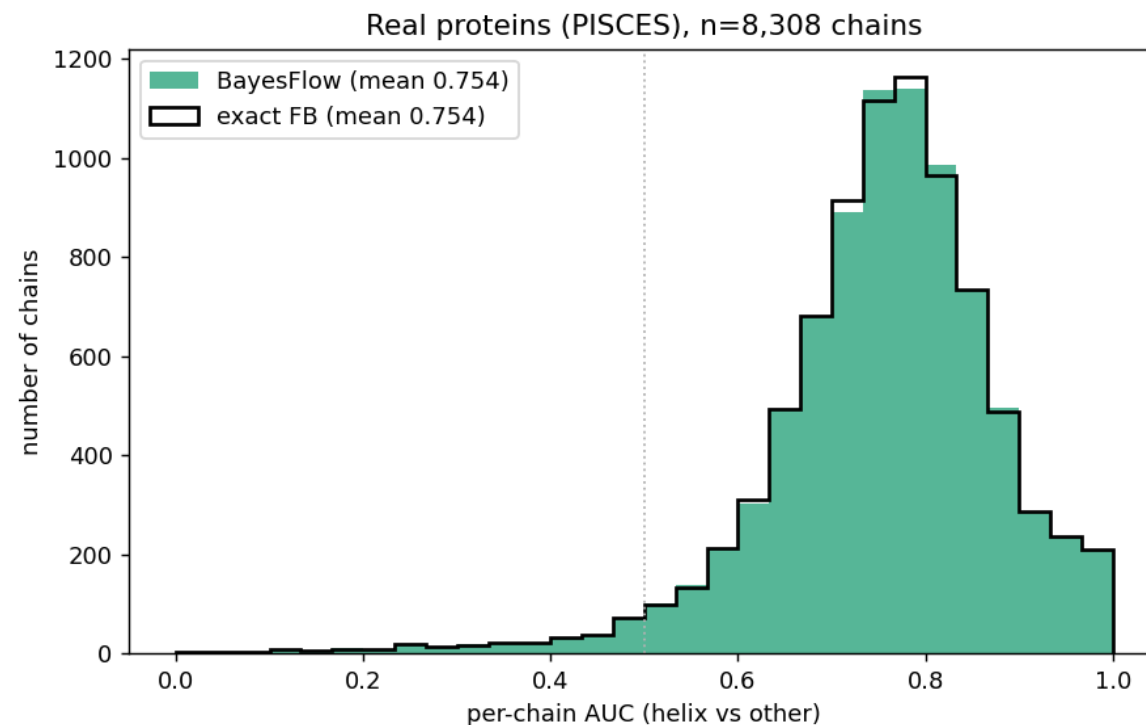


Fig 6. Per-chain AUC over 8,308 real chains (those with both classes).

Result 1: the neural posterior reproduces exact inference

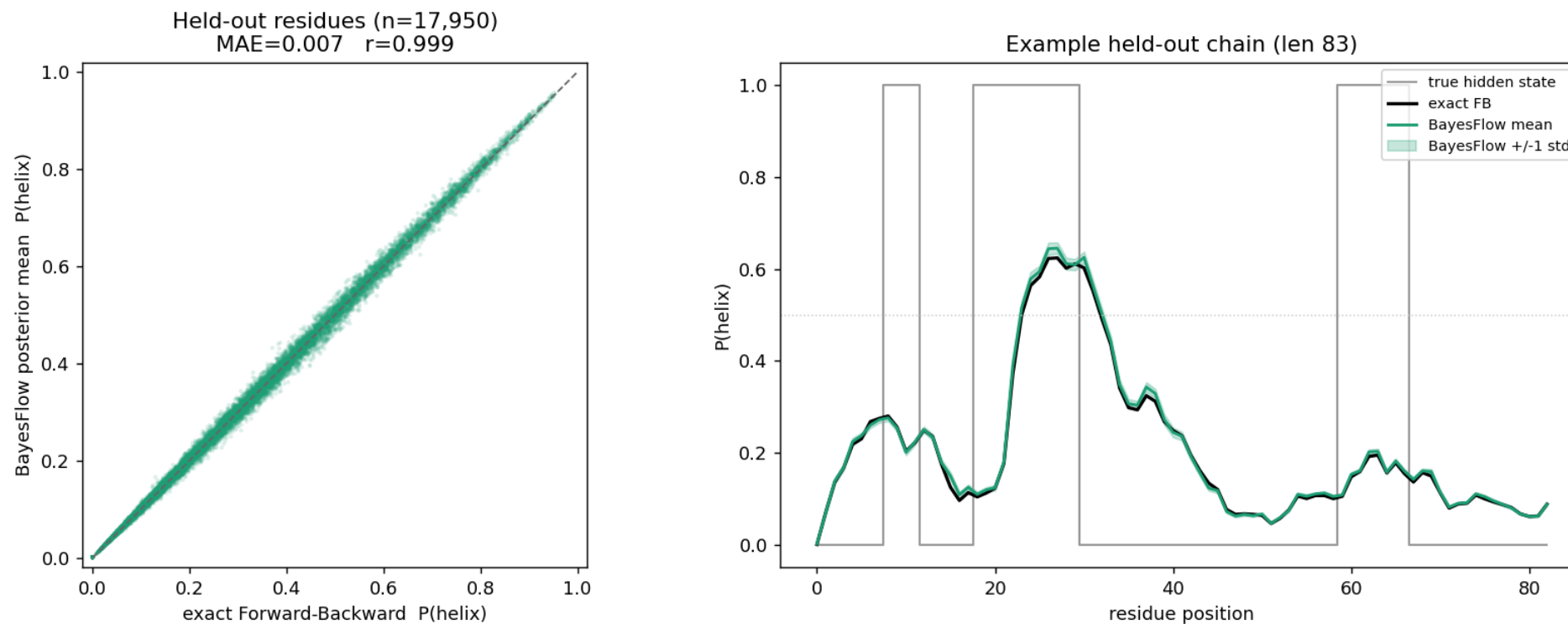


Fig 7. Left: BayesFlow mean vs exact FB ($r=0.999$). Right: an example chain with the ± 1 std band.

Held-out simulated chains: correlation 0.999, MAE 0.007. The fast neural posterior is interchangeable with slow HMM Forward–Backward.

SBI diagnostics: convergence, calibration, contraction

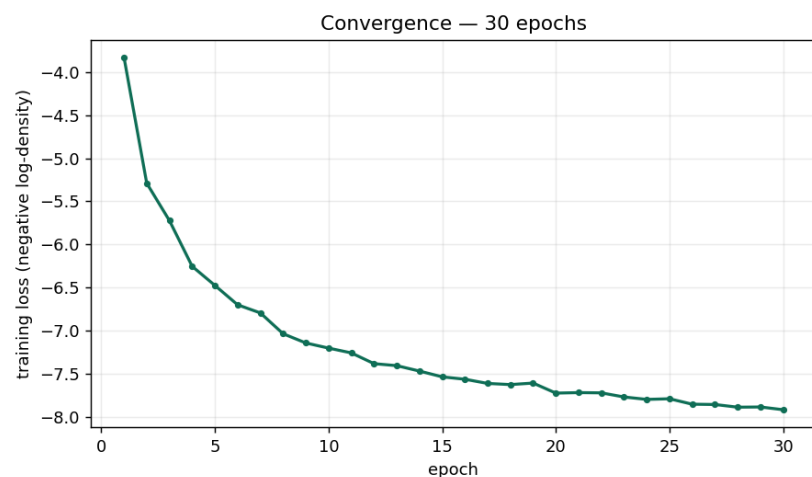


Fig 8a. Training loss.

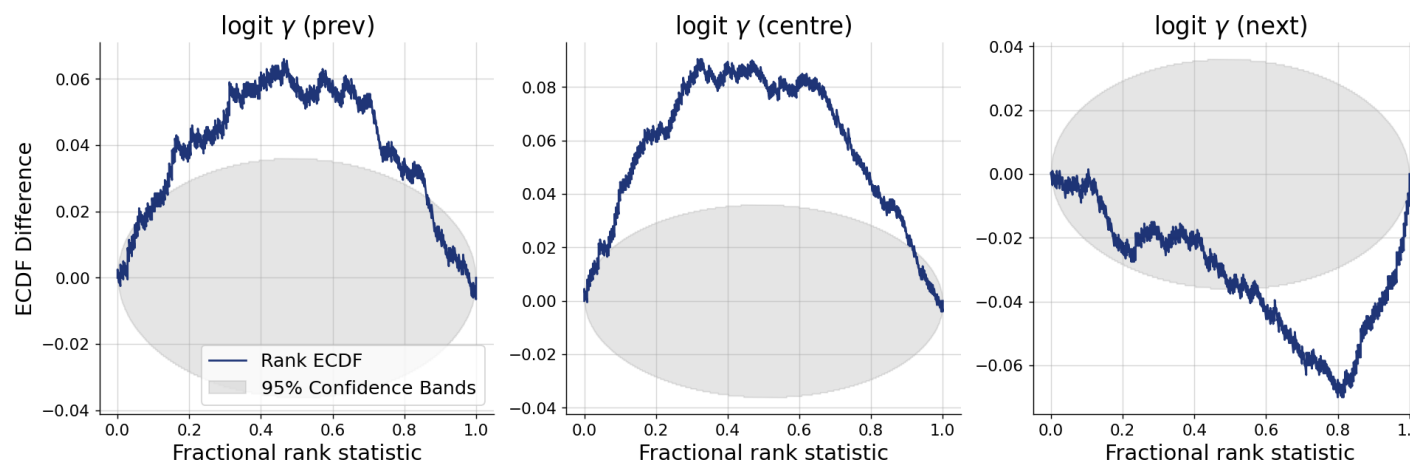


Fig 8b. Simulation-based calibration (SBC) rank ECDF, per target dim.

- Convergence: loss $-3.8 \rightarrow -7.9$, plateaued. Recovery $r = 0.999$ and posterior contraction 0.999 — the estimate is excellent and tight.
- SBC (right): the rank ECDF slightly exits the 95% band $\rightarrow$ a small location bias (~ 0.15 posterior SD ≈ 0.002 in probability). The uncertainty band is not perfectly calibrated.
- Honest limitation: point estimates are unaffected; likely cause is the near-degenerate 3-D target (dim-correlations 0.89–0.96) vs a coupling flow. Candidate fix left as future work.

Results at a glance

Setting	AUC BayesFlow	AUC exact FB	Acc@0.5 BayesFlow	Acc@0.5 exact FB	Majority baseline
Held-out simulated	0.798	0.798	0.760	0.760	0.675
Insulin B (wild-type 1MSO)	0.981	0.986	0.633	0.633	0.633
Insulin A (wild-type 1MSO)	0.519	0.509	0.429	0.429	0.571
Real PISCES (8,994 chains)	0.754	0.754	0.743	0.743	0.678

Table 1. AUC & accuracy@0.5 vs the majority-class baseline — BayesFlow vs exact FB.

- BayesFlow $\approx$ exact FB in every setting — faithful amortization.
- Accuracy@0.5 only beats the trivial baseline on the calibrated (simulated / pooled real) settings; on insulin it ties or falls below it $\rightarrow$ AUC is the fair metric.
- FB is a ceiling on SIMULATED data only; on real proteins the HMM is misspecified, so a better model can beat it.

TL;DR — take-home message

- α -helix as a 2-state HMM whose transition AND emission tables match real proteins ($r \approx 0.99$).
- BayesFlow amortized posterior trained on exact Forward–Backward targets via a sliding 31-residue window; it reproduces exact FB almost perfectly ($r = 0.999$).
- Diagnostics: excellent recovery & contraction, but SBC reveals a small (~ 0.15 SD) calibration bias — a real, owned limitation.
- On unseen real proteins it ranks propensity-driven helices well (insulin B-chain 0.98; PISCES 0.75) but fails where structure is 3-D-stabilised (insulin A-chain 0.52) — the price of a sequence-only model.

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